

Stable Temporal Retrieval with Direct Supersession Certificates

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Abstract

Archives contain relevant statements that describe different periods. Temporal retrieval must select evidence that applies to the requested date. An incorrect connection between claims can also change earlier evidence when a system merges their histories. We study this propagation problem through direct supersession certificates. Each certificate stores the first later claim accepted as a replacement by a supplied pairwise matcher. We characterize this operation when matching decisions do not form consistent groups. One certificate per claim reproduces replacement decisions at every requested date. Later-dated additions preserve earlier eligibility decisions. Changing one pair affects at most one claim interval. With a fixed ranking, the selected set changes by at most one removal and one insertion. An exact compiler produces these certificates without gold entity or value annotations. Component timelines change past eligibility at 8 of 11 TempLAMA event times and 27 of 42 Wikipedia event times. Direct certificates have no such changes. Complete TempLAMA answer sets give 94.7% current evidence accuracy, compared with 88.3% for exact-group date selection. Historical stale-context exposure falls from 16.6% to 1.8%, with historical accuracy of 98.2% versus 98.4%. The results connect formal stability to source-attributed evidence selection.

1 Introduction

An archive can contain several correct descriptions of an attribute from different periods. A company appoints a new director. A product changes its supported features. A policy replaces an earlier threshold. Retrieval-augmented generation must select the description that applies to the question’s date (Lewis et al., 2020). Text similarity identifies the topic, but does not establish that period.

Consider an illustrative organization, Northstar. A 2019 record names Mira as its director. A 2021

record names Leon. A 2024 record names Sana. A current question needs Sana’s record. A question about 2022 needs Leon’s record. Both questions search the same archive. Deleting older records prevents historical answers. Keeping all relevant records leaves the reader to resolve competing states within a limited context. The required operation is temporal evidence selection.

One approach groups related claims and orders each group by time. An earlier statement then expires when another value appears in its group. This works when every group corresponds to one coherent history. Text comparisons, however, can connect claims from different subjects or attributes. If grouping uses connected components, one incorrect connection can merge their histories. Time ordering can then treat unrelated statements as replacements. The resulting exclusions can extend beyond the mistaken pair. They can also change answers about periods before either bridge endpoint existed.

We ask a structural question: *How can a retrieval memory support historical states while limiting the propagation of comparison errors?* We study direct supersession certificates. A certificate connects a claim to its earliest accepted replacement. Its start and endpoint determine when that claim may enter the reader’s context. Each certificate preserves the source of the replacement decision. Figure 1 shows the complete selection process.

Direct temporal invalidation already appears in Graphiti (Rasmussen et al., 2025). Our contribution is its formal behavior under arbitrary pairwise decisions. The matcher can make errors. Two accepted links need not justify a third link. We prove that one certificate per claim exactly represents every temporal cutoff. We also prove that later-dated additions preserve earlier eligibility decisions. Finally, changing one pair can affect only its older endpoint’s interval. Under fixed relevance scores,

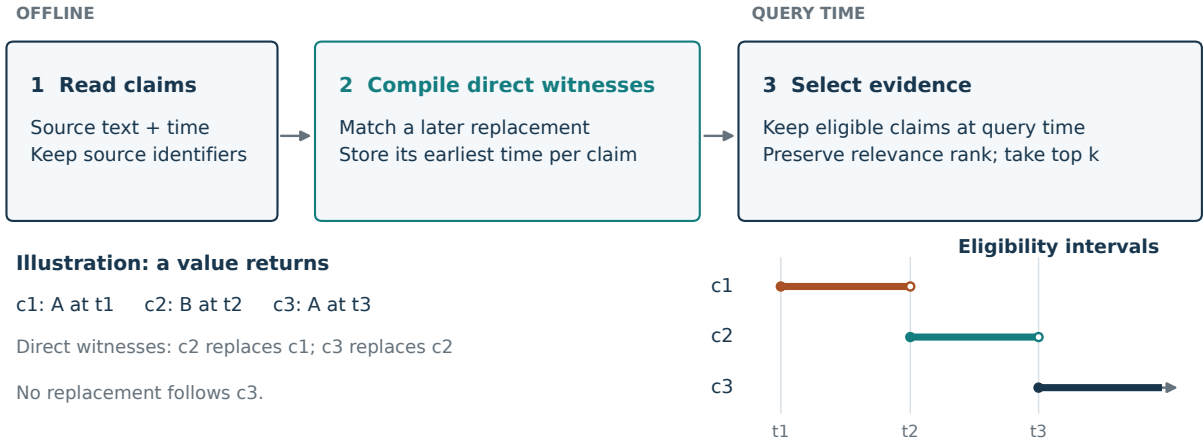


Figure 1: Direct certificates connect evidence selection to its replacement sources. The compiler stores the first accepted replacement for each claim. Query-time filtering preserves relevance order before selecting the context. The lower illustration shows a value that returns after replacement. Each occurrence keeps its own interval. Filled endpoints are included; open endpoints are excluded.

the selected set changes by at most one removal and one insertion. These properties follow from direct dependence on replacement evidence. They do not require perfect entity resolution.

We implement an exact compiler that scans timestamp batches and retires each claim after finding its first witness. The compiler receives text, identifiers, and timestamps. It preserves the supplied matcher while avoiding transitive grouping. Complete candidate blocking reduces unnecessary comparisons without changing the output.

Our evaluation separates stability from retrieval accuracy. It tests observed histories, individual pair changes, full-pool evidence selection, and compiler equivalence. We also inspect the complete answer sets behind TempLAMA (Dhingra et al., 2022). This inspection distinguishes a changed selected answer from an actual removal of the earlier answer. The resulting protocol evaluates all source-supported answers with unchanged rankings.

The contributions are a formal account of direct invalidation, an exact compiler, and an empirical analysis of temporal error propagation. The experiments compare direct certificates with component timelines and exact-group date selection under shared extraction and relevance scores. This design identifies which benefits arise from compilation and which still depend on the matcher.

2 Related Work

Temporal memory. Graphiti identifies contradictions and closes a fact’s interval using a later contradictory fact (Rasmussen et al., 2025). Its im-

plementation includes earliest-time direct invalidation (Graphiti contributors, 2026). MemStrata applies deterministic supersession to structured keys with one mutable value (Yadav, 2026). These systems establish direct invalidation and retained history as existing mechanisms. We analyze their underlying compilation problem when pair decisions need not define consistent groups. Our results concern all-cutoff equivalence, temporal prefix consistency, and the influence of changed pair decisions.

Re³ extracts and groups facts from retrieved documents (Cao et al., 2026). Its recency filter suppresses conflicting values using publication dates. It also learns temporal relevance. Our study holds extraction and relevance fixed while comparing ways to compile replacement decisions. Reddy and Challaram similarly separate evidence extraction from policy execution (Reddy and Challaram, 2026). Their analysis motivates comparisons that do not attribute better extraction to a different temporal executor.

Temporal retrieval and reasoning. TempRALM combines semantic scores with query–document time distance (Gade et al., 2025). Half-life scoring combines semantic relevance with a recency prior (Grofsky, 2025). These methods select evidence through scores rather than explicit replacement witnesses. TA-RAG retrieves evidence for diachronic questions (Lau et al., 2025). IA-RAG represents interval events and uses interval relations for temporal reasoning (Wang et al., 2026). Its scope includes interval extraction and refinement. Our certificates instead compile

endpoints from supplied pair decisions. Temporal knowledge graphs also model dated relational evidence (Trivedi et al., 2017; Lacroix et al., 2020). The relevant distinction here is between direct decisions and closure over inferred claim groups.

Temporal evaluation. TimeQA, TempLAMA, and StreamingQA study knowledge that changes over time (Chen et al., 2021; Dhingra et al., 2022; Liška et al., 2022). RealTime QA evaluates current questions, while FreshLLMs refreshes evidence through web search (Kasai et al., 2023; Vu et al., 2024). FRESCO includes obsolete passages and current but insufficient passages (An et al., 2026). It tests temporal evidence selection beyond global newest-document ranking. Our experiments add two structural measurements: historical prefix disagreement and the effect of one changed pair decision. We report retrieval metrics separately from archived reader outputs.

Structured retrieval and conflicting context. GraphRAG and HippoRAG use graph structure for evidence organization and retrieval (Edge et al., 2024; Gutiérrez et al., 2024). Their graph structures serve broader retrieval goals than the compiler studied here. Studies of conflicting context show that answer generation depends on the evidence presented to the reader (Longpre et al., 2021; Xie et al., 2024). Context position also affects language-model behavior (Liu et al., 2023). These findings support measuring the selected evidence itself. Our stability bound applies before generation and requires no assumption about reader behavior.

3 Direct Supersession Certificates

3.1 Inputs and temporal eligibility

Let $\mathcal{C}$ contain source-attributed claims. A dated claim c has text x_c , identifier id_c , and finite timestamp t_c . The timestamp identifies the supplied source state or recorded change time. A publication date alone does not establish an unobserved effective date. The query supplies a cutoff τ and a relevance ranking π_q over the corpus. Our output contains at most k claims that remain eligible at τ .

The pairwise gate $G(d, c)$ means that d supplies a replacement for c . It combines a shared-attribute decision with a changed-value decision. The gate uses fixed claim features or a fixed pairwise procedure. It need not be transitive, symmetric, or correct against world knowledge. We study the evidence policy induced by this gate. Interpreting its

output as factual validity additionally requires correct replacement decisions and correctly ordered change times. Concurrent values require a gate that distinguishes coexistence from replacement.

Define eligible evidence by

$$A_G(\tau) = \{c : t_c \leq \tau, \nexists d \in \mathcal{C} \text{ such that } t_c < t_d \leq \tau \text{ and } G(d, c)\}. \quad (1)$$

Thus a claim remains eligible until an accepted later replacement becomes available at the cutoff. Equal timestamps do not establish replacement order. Undated claims receive no historical certificate. The current-answer interface retains them as unresolved evidence, matching the archived system’s policy.

3.2 A certificate for every cutoff

For each dated claim, define its first replacement boundary:

$$b_G(c) = \min\{t_d : t_d > t_c, G(d, c)\}, \quad (2)$$

with $\min \emptyset = \infty$. Store a witness $w_G(c)$ that attains this boundary. The smallest identifier selects among witnesses with the same timestamp. The certificate is

$$I_G(c) = [t_c, b_G(c)). \quad (3)$$

The interval determines eligibility. The witness identifies the source responsible for its endpoint. An infinite endpoint means that the corpus contains no accepted later replacement.

The selected context is

$$R_G(q, \tau) = \text{Top}_k\langle c \in \pi_q : \tau \in I_G(c) \rangle. \quad (4)$$

Filtering precedes the context budget. The operation preserves the relative relevance order of retained claims. It does not fill empty slots with excluded claims. Historical and current retrieval use the same certificates. For dated claims, the final current mask retains exactly those with infinite endpoints.

Repeated values remain separate source records. A return to an earlier value creates a new record with its own start time. The earlier occurrence retains its original endpoint. The method therefore represents recurrence without reopening an expired source record. For the Northstar illustration, each appointment closes the preceding record through its own direct decision.

Inputs: dated claims, fixed gate G , complete postings.
Output: boundary $b(c)$ and witness $w(c)$.

1. Set $b(c) = \infty$ and $w(c) = \emptyset$ for every claim.
2. Initialize the unresolved index $U = \emptyset$.
3. Process timestamp batches B in increasing order.
4. For each $d \in B$, process identifiers in order.
5. Find unresolved candidates c from complete postings.
6. If $G(d, c)$, set $(b(c), w(c)) = (t_d, d)$.
7. Remove each resolved c from all postings in U .
8. After all comparisons for B , insert its claims into U .

Figure 2: Exact compilation. Steps 4–7 run inside each timestamp batch. Removing resolved claims avoids repeated work. Delayed batch insertion prevents replacement between simultaneous claims.

3.3 Exact compilation

Figure 2 gives the compiler. It processes timestamp batches in increasing order. An inverted index stores unresolved older claims. Each new claim queries complete candidate postings and evaluates the gate against unresolved candidates. The first accepted replacement fixes an older claim’s endpoint. The compiler then removes that older claim from the postings. No later replacement can improve its earliest endpoint. Claims from the current batch enter the postings only after all comparisons finish. This preserves equal-time claims and deterministic witness ties.

For the supplied text matcher, a positive frame threshold permits complete token blocking. Accepted nonempty frames share a frame token. All empty frames use one common posting key. A nonpositive threshold uses one global posting key. Custom gates use a full unresolved scan unless their candidate index has an equivalent completeness guarantee. Blocking changes the comparisons examined, but not the certificates produced.

The supplied gate uses the earlier detector’s fixed parsing rules. It compares relation frames, named phrases, and changed numeric or textual values. We preserve these rules throughout the compiler comparisons. No gold subject, relation, answer value, or chronology enters the compiler. Section 5 identifies the annotations used only to construct questions and score results.

Let $n = |C|$. Let L count posting entries, V

posting visits, and P unique pair tests. If one gate evaluation costs at most g , construction costs

$$O(n \log n + L + V + Pg) \quad (5)$$

after feature extraction. Output storage is $O(n)$ words. Uninformative postings and long repeated-value runs can still require quadratic work. We therefore report pair tests and posting visits separately from elapsed time.

4 Exactness and Error Propagation

The analysis treats the supplied gate as an arbitrary fixed relation. It separates compiler correctness from the semantic accuracy of its inputs. Complete proofs and boundary cases appear in Appendix A.

Theorem 1 (All-cutoff equivalence). *For every dated claim c and finite cutoff τ ,*

$$c \in A_G(\tau) \iff \tau \in I_G(c). \quad (6)$$

At most one stored witness per claim therefore represents every temporal eligibility decision.

Proof. If $t_c > \tau$, both statements are false. Otherwise, an accepted replacement exists by τ exactly when $b_G(c) \leq \tau$. Taking the complement gives the equivalence. $\square$

This result also explains why a newest replacement is insufficient for historical retrieval. A newest witness establishes eventual replacement, but can miss an earlier endpoint. Both choices nevertheless give the same final current mask. Each excludes exactly the claims with at least one accepted later witness.

Corollary 1 (Temporal prefix consistency). *Let $C_{\leq \tau} = \{c : t_c \leq \tau\}$. Querying full-corpus certificates at τ equals compiling the final mask on $C_{\leq \tau}$.*

Proof. Both procedures examine precisely the accepted pairs satisfying $t_c < t_d \leq \tau$. $\square$

Appending records dated after τ therefore preserves the evidence state at τ . This statement concerns timestamp order, not file arrival order. A newly received record with an earlier timestamp can legitimately change the corresponding historical state. The gate must also remain fixed when the corpus grows.

Theorem 2 (Pair-local influence). *Fix claim timestamps and identifiers. Suppose gates G and H differ on h temporally ordered pairs. Let U contain the older endpoints of those pairs. For every cutoff,*

$$A_G(\tau) \triangle A_H(\tau) \subseteq U, \quad |U| \leq h. \quad (7)$$

With a fixed ranking and deterministic ties,

$$|R_G(q, \tau) \triangle R_H(q, \tau)| \leq 2 \min\{k, |U|\}. \quad (8)$$

Proof. A claim’s boundary depends only on pairs whose older endpoint is that claim. Its interval cannot change outside U . Changing one eligibility bit removes or adds at most one selected claim and one replacement. Apply this argument successively over U . The two contexts also contain at most $2k$ claims altogether. $\square$

For one changed pair, the interval difference is confined to one record. Its temporal extent is also explicit. Writing $J_c = [\min\{b_G(c), b_H(c)\}, \max\{b_G(c), b_H(c)\}]$, we have

$$c \in A_G(\tau) \triangle A_H(\tau) \iff \tau \in J_c. \quad (9)$$

For a query-time distribution μ , the expected mask difference equals $\sum_c \mu(J_c)$. This identifies the period affected by each changed decision. The bound concerns pair decisions, so a parser update can affect several intervals through several changed pairs.

Why component closure differs. An incorrect bridge can change intervals throughout a connected component. Consider one attribute with repeated value a at times $0, \dots, m-1$. It changes to b at time $2m$. A second attribute has z at time m and w at time $2m+1$. All four values differ. An incorrect accepted pair between b and w merges the histories. Component sorting then closes every earlier a record at time m . At cutoff $m + \frac{1}{2}$, it excludes m records through one changed pair. Direct compilation changes none at that cutoff because the affected older endpoint has not appeared. This construction gives unbounded component amplification and a failure of temporal prefix consistency. It is a correctness example, not a benchmark observation.

Proposition 1 (Compiler exactness). *The batched compiler returns Equation 2 when its candidate postings contain every pair accepted by the gate.*

Corpus	Claims	Query keys	Past	Current
Wikidata	208	33	125	33
TempLAMA	747	300	447	300
Wikipedia prose	132	35	97	35

Table 1: Evaluation pools and query counts. Past queries use distinct adjacent source dates. TempLAMA uses complete source answer sets for scoring.

Proof. Before each batch, the index contains exactly the earlier claims without an accepted earlier witness. Completeness exposes every accepted pair from the batch to those claims. The first such batch supplies the minimum replacement time. Removing resolved claims cannot change that minimum. Inserting after comparison prevents equal-time replacement. $\square$

5 Experimental Protocol

We use three evidence collections with distinct source forms. Table 1 records their pool sizes and query counts. No synthetic benchmark contributes to the main results. Constructed examples test implementation correctness only.

Wikidata. The archive contains 208 rendered claims over 73 annotated attribute keys (Vrandečić and Krötzsch, 2014). Thirty-three changing keys produce current and historical queries. Timestamps come from recorded starts. The corpus also contains stable attributes that serve as distractors. All claims share one retrieval pool.

TempLAMA and complete answer sets. The released adaptation contains 747 rendered claims over 300 keys (Dhingra et al., 2022). Its builder selects the first listed answer at each year and collapses repeated selected answers. This produces a selected-answer transition task. We recovered all 2,677 original yearly records for these keys. Of 447 selected-answer changes, 153 leave the previous answer in the new complete answer set. Moreover, 245 keys have multiple supported answers in at least one year. Therefore, our primary TempLAMA scoring accepts every answer in the original source set.

Rankings, gates, and the 747-claim pool remain fixed during this source-set rescore. At each cutoff, the latest observed yearly source set defines valid answers. That set persists between observations and after the final observation within this snapshot evaluation. The experiment has 447 historical

probes and 300 current probes. A separate source-defined subset contains 55 keys with exactly one answer in every observed year. It retains the full retrieval pool and has 57 historical probes. Subset selection uses source metadata alone.

RealProse. This collection contains 132 genuine Wikipedia lead extracts over 35 subjects and five relation families. The families cover corporate executives, government heads, heads of state, club managers, and organization leaders. Curated tenure starts supply timestamps. The resulting evaluation has 97 historical and 35 current probes. These are source extracts with annotated times, rather than archived page revisions. The CEO subset is reported only as a breakdown, since it overlaps RealProse.

Visible inputs and queries. Inference receives new claim objects containing only identifiers, text, and timestamps. Gold keys and values are removed before matching, compilation, and ranking. Benchmark templates or subject–relation labels generate questions without answer values. Historical probes use midpoints between distinct observed dates within each query key. Current probes follow the corpus’s latest date. Every method searches the same full pool. No gold claim selects a timeline or seeds retrieval.

Matched baselines. All new retrieval conditions use the same BM25 ranking (Robertson et al., 1995). We fix $k_1 = 1.5$, $b = 0.75$, and the context budget at five claims. All methods exclude records dated after the query cutoff. *BM25* adds no replacement filter. *Date-top5* reorders five BM25 candidates by date. *Date-full-pool* ranks the entire available pool by date, with relevance breaking date ties. *Exact-group latest* groups identical extracted frames and subject phrases, then retains the latest available records per group. All records tied at that date survive.

Text-CC links each claim to its newest accepted replacement, then sorts connected claims into intervals. It uses extracted value features, without gold values. This matched text-based control replaces the archived module’s access to annotated values. *Direct latest* uses the last accepted replacement as the endpoint. It tests whether retaining a replacement witness alone is sufficient. *Direct first* implements our certificates. The Graphiti invalidation primitive has the same endpoint semantics under identical ordered pair decisions. Our matched-gate

Corpus	Cutoffs	Closure	Direct
Wikidata	27	0	0
TempLAMA	11	8	0
Wikipedia prose	42	27	0

Table 2: Cutoffs where future-dated records change past eligibility. Both constructions use the same frozen pair rules. Zero denotes prefix consistency.

comparison concerns that primitive, rather than a separately executed Graphiti pipeline.

Measurements. Evidence Hit@1 requires a returned claim to match the requested key and a value valid at the cutoff. Hit@5 requires at least one such claim among five. Stale@5 measures whether any selected claim carries a value invalid at its own key’s cutoff state. This metric can reward an irrelevant but current context, so we report it alongside Hit@1 and Hit@5. Paired confidence intervals resample complete key clusters with 5,000 repetitions and seed 20261006.

Prefix tests compare full-corpus intervals against intervals compiled from claims available at each observed event time. Pair-sensitivity tests remove each observed accepted pair individually and rebuild both compilers. They record the number of affected intervals and membership changes across event times. Gold keys classify cross-key decisions only after these interventions. Finally, exhaustive pair scans test compiled boundaries, witness ties, and every distinct eligibility interval.

6 Results

6.1 Later records can change earlier component states

Table 2 tests temporal prefix consistency using observed records. Component intervals disagree with prefix-only compilation at 8 of 11 TempLAMA event times. They disagree at 27 of 42 RealProse event times. Direct certificates have zero disagreements on both collections. Wikidata has no disagreement under either compiler. These results identify a structural failure that depends on the observed pair graph.

The differences can reach the retrieved evidence. Full-corpus versus prefix-only component compilation changes the first result for six TempLAMA historical probes and eight RealProse probes. The RealProse versions have equal aggregate Hit@1 despite those different selections. Thus aggregate accuracy alone does not detect this instability. The

Corpus	Pairs	Closure	Direct
Wikidata	452	4	1
TempLAMA	698	3	1
Wikipedia prose	273	8	1

Table 3: Maximum distinct claims affected by deleting one observed positive pair. Counts include valid and erroneous pairs. All event cutoffs are checked.

certificate guarantee applies at each cutoff and record, rather than only to an average score.

6.2 One pair can affect several component intervals

Table 3 removes accepted pairs from the observed corpora. Direct compilation changes at most one interval in every intervention. The corresponding maxima for component compilation are four on Wikidata, three on TempLAMA, and eight on RealProse. The intervention set contains both correct and incorrect pairs. It measures sensitivity to a changed decision, rather than treating every removed pair as an error correction.

Cross-key pairs provide a separate error-focused analysis. TempLAMA contains 46 accepted cross-key pairs; RealProse contains 96. Removing one such pair changes up to three component intervals in either collection. Across RealProse event times, these corrections produce 68 component membership changes and 37 direct membership changes. Several pair changes leave direct endpoints unchanged because another witness establishes the same first boundary. The measured influence agrees with Theorem 2.

6.3 Evidence accuracy and stale exposure

Table 4 reports full-pool retrieval with matched relevance scores. Its main TempLAMA rows use complete original answer sets. Direct certificates reach historical Hit@1 of 0.982, compared with 0.781 for BM25. Exact-group latest reaches 0.984 and component intervals reach 0.975. The direct-component difference is 0.0067, with a 95% interval of $[-0.0023, 0.0167]$. The direct-exact-group difference is -0.0022 , with interval $[-0.0183, 0.0153]$. These close accuracy results accompany different evidence exposure.

Historical Stale@5 is 0.709 for BM25, 0.166 for exact grouping, and 0.018 for direct certificates. Component intervals also reach 0.018. Direct certificates preserve historical Hit@5 of 0.991, while exact grouping reaches 1.000. The comparison

therefore measures a tradeoff between retaining valid alternatives and excluding obsolete evidence. For current TempLAMA queries, direct and component selection reach Hit@1 of 0.947. Exact grouping reaches 0.883. Their difference is 0.0633, with interval $[0.0333, 0.0933]$.

RealProse tests the same operations on genuine prose. Historical Hit@1 rises from 0.567 for BM25 and exact grouping to 0.835 for direct and component intervals. The direct gain over exact grouping is 0.268, with interval $[0.161, 0.378]$. Direct Hit@5 is 0.938, compared with 0.918 for component intervals. Their Stale@5 values are 0.485 and 0.381, respectively. The direct rule preserves some evidence that component closure removes. Its bounded influence does not imply lower stale exposure in every corpus.

On the source-defined single-answer subset, direct certificates recover 55 of 57 historical first results. Exact grouping recovers all 57; component intervals recover 54. Direct and component Stale@5 are both 0.018, compared with 0.193 for exact grouping. This subset confirms that stale exposure and first-result accuracy measure different properties.

6.4 Exact compilation and current-mask preservation

The compiler matches exhaustive direct evaluation on all packaged corpora. The checks cover endpoints, selected witnesses, eligibility cutoffs, timestamp ties, and missing timestamps. They also verify exact equality with the archived current mask. Table 5 reports the main corpora’s comparison counts.

For Wikidata, the indexed compiler tests 2,738 pairs against 19,981 strictly ordered pairs in the exhaustive scan. For TempLAMA, the counts are 26,538 and 241,154. For RealProse, the indexed compiler tests 2,876 pairs. These reductions preserve the full set of earliest boundaries. They measure this matcher and these corpora, while Equation 5 retains the general worst-case cost.

The preserved current masks also preserve previously generated contexts under the archived ranking. Appendix C records the inherited reader results and their original scoring task. Those runs assess the unchanged current-answer path. The new results in this section assess compiler behavior and evidence selection.

Method	Historical ($n = 447$)		Current ($n = 300$)	
	Hit@1 $\uparrow$	Stale@5 $\downarrow$	Hit@1 $\uparrow$	Stale@5 $\downarrow$
BM25	78.08	70.92	23.33	100.00
Date within top five	54.36	70.92	60.67	100.00
Global date	28.41	0.00	12.67	0.00
Exact-group latest	98.43	16.55	88.33	28.33
Text-CC intervals	97.54	1.79	94.67	16.33
Direct certificates	98.21	1.79	94.67	16.33

Table 4: TempLAMA evidence selection, in percent. Complete yearly answer sets define correct and stale values. All methods use the same 747 claims. Direct certificates preserve the fixed BM25 ranking after filtering. These are retrieval results, not generated-answer scores.

Corpus	All pairs	Compiler	Ratio
Wikidata	19,981	2,738	7.30
TempLAMA	241,154	26,538	9.09
Wikipedia prose	8,432	2,876	2.93

Table 5: Pair tests for exhaustive and indexed construction. The ratio measures work reduction, not wall-clock speed. Outputs match the exhaustive oracle exactly.

7 Discussion

What the guarantee adds. A direct certificate gives each exclusion a specific source and start time. The local-influence theorem shows how far a changed replacement decision can propagate. The prefix theorem additionally controls which periods it can affect. These properties support debugging and incremental maintenance. An operator can inspect the witness responsible for a missing passage without reconstructing an entire component. With a fixed candidate ranking, a single corrected pair requires reconsidering one eligibility bit at the query cutoff.

Why accurate grouping can still be sufficient. Exact-group date selection is a strong baseline when extracted keys remain stable. Its historical TempLAMA accuracy matches or exceeds the direct construction in several comparisons. Direct certificates become useful when pair decisions carry information that exact equality misses. They preserve that information without assuming transitivity. The RealProse result illustrates the benefit of approximate matching. The prefix and sensitivity results identify the additional cost of converting those matches into components.

From replacement to coexistence. The TempLAMA source audit reveals that answer order cannot establish replacement. A previous employer, team, or office can remain valid when another value appears. Complete answer sets expose this distinc-

tion during evaluation. A deployment gate should also express it during inference. For instance, evidence of a new concurrent role should not automatically close an existing role. The compiler can accept such a gate unchanged. Its formal guarantees then apply to that gate’s decisions.

Practical integration. The certificate layer sits between relevance ranking and context selection. It can therefore reuse an existing retriever and reader. The source text remains available for citation and historical retrieval. When an application’s metadata supplies actual effective dates, those dates define the applicable periods. When it supplies publication dates, the certificates instead describe the policy over those dated records. The application must choose that interpretation before compiling its memory. The same choice determines what a dated user question asks the system to retrieve.

8 Conclusion

Direct supersession certificates give temporal evidence selection a precise dependence on pairwise replacement decisions. One witness per claim preserves every cutoff, while later-dated additions leave earlier evidence states unchanged. The pair-local bound connects a matching correction to its affected intervals and selected context. Observed prefix failures and pair interventions show why this distinction matters for component-based timelines. The resulting compiler supports historical retrieval with explicit sources for each exclusion.

Limitations

The certificates are exact relative to supplied pair decisions and timestamps. They do not verify source truth or infer effective dates from publication metadata. Concurrent values require replacement decisions that distinguish coexistence from retirement. Our current matcher does not solve that general extraction problem.

The new experiments use small English collections and a fixed lexical ranking. Wikidata and TempLAMA claims are rendered from structured records. RealProse contains genuine extracts but curated tenure dates. Its 35 subjects provide limited coverage of open-domain prose. The complete TempLAMA answer-set analysis is a development evaluation with frozen rankings and no subsequent parameter tuning. Its retrieval pool still contains only the previously rendered selected answers. It therefore does not evaluate exhaustive retrieval of every valid concurrent answer.

Direct invalidation is an existing temporal-memory operation. Our comparison formalizes its behavior under shared pair decisions. The experiments compare matched compilation rules, rather than complete Graphiti, MemStrata, or Re³ pipelines. The archived reader runs evaluate an unchanged current-answer path. New historical results measure retrieved evidence rather than generated answers.

The stability theorem fixes relevance scores and timestamps. One document edit can change several pair decisions or alter the ranking. One changed evidence item can also change a generated answer. The certificate guarantee bounds structural propagation, rather than downstream answer sensitivity.

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A Proofs and Boundary Cases

A.1 All-cutoff equivalence

Fix a dated claim c and a finite cutoff τ . If $\tau < t_c$, the claim has not appeared in either representation. Otherwise define the witness set

$$W_\tau(c) = \{d : t_c < t_d \leq \tau, G(d, c)\}.$$

This set is nonempty exactly when its earliest possible timestamp satisfies $b_G(c) \leq \tau$. Thus $W_\tau(c) = \emptyset$ exactly when $t_c \leq \tau < b_G(c)$. This proves Theorem 1, including the empty-witness case. Equal timestamps never belong to $W_\tau(c)$. The identifier tie rule changes the stored source, but not eligibility.

For a prefix $C_{\leq \tau}$, the relevant witness set is exactly $W_\tau(c)$. Claims outside the prefix cannot appear in it. This proves Corollary 1 without requiring correct or transitive pair decisions. It requires fixed features and a fixed gate. Corpus-dependent feature changes constitute different gates and belong to the influence analysis instead.

A.2 Pair-local influence and temporal footprint

Let $D = \{(d, c) : t_d > t_c, G(d, c) \neq H(d, c)\}$. Let $U = \{c : \exists d, (d, c) \in D\}$. For each $c \notin U$, every possible witness has the same gate value under both compilers. Therefore $b_G(c) = b_H(c)$. Its certificate and eligibility agree at every cutoff. This proves the mask containment and the bound $|U| \leq |D| = h$.

For the context bound, change the eligibility bits in U one at a time. Adding one eligible record either leaves the selected context unchanged or inserts it at its fixed ranking position. At capacity, that insertion displaces one record. Removing one eligible record either leaves the context unchanged or removes that record. The next eligible record can fill its slot. Each bit change therefore contributes at most two elements to the symmetric difference. The triangle inequality gives $2|U|$. Since each context has size at most k , the bound is also at most $2k$.

Both certificates start at t_c . Their indicator functions differ exactly between their endpoints. Hence Equation 9 follows, including an infinite endpoint. By linearity of expectation,

$$\mathbb{E}_{\tau \sim \mu} |A_G(\tau) \triangle A_H(\tau)| = \sum_c \mu(J_c).$$

The context expectation is at most twice this quantity. The statement uses the same ranking at each compared cutoff. It does not require independent claim errors or independent query times.

A.3 Amplification through a component bridge

Use $m \geq 2$ records with value a at times $0, 1, \dots, m-1$. Their shared attribute changes to b at time $2m$. A second attribute contains z at time m and w at time $2m+1$. Assume a, b, z, w are distinct. Initially all matching decisions are correct. Each a record has a replacement link to b . The z record has a replacement link to w .

Add the single false decision that w replaces b . The component graph now joins both histories. Sorting this component makes z the first later distinct value for every a record. Their component endpoints become m . At cutoff $m + \frac{1}{2}$, all m records therefore become ineligible. The direct construction changes only b 's boundary. Because b starts after the cutoff, direct eligibility does not change there.

The prefix at this cutoff contains neither endpoint of the false bridge. Consequently, component reconstruction on that prefix cannot make the same exclusions. This gives a prefix inconsistency caused entirely by later-dated records. For a context difference, add k independent filler records and rank the a records first. When $k \leq m$, direct selection returns k such records. Component selection replaces them with the fillers. The contexts have symmetric difference $2k$. This construction establishes possibility, not an observed frequency.

A.4 Exact indexed compilation

Order timestamp batches increasingly. Within a batch, order candidate witnesses by identifier. Before processing the first batch, the unresolved index is empty. Assume it contains precisely earlier claims without an accepted earlier witness before some batch. Complete posting lists expose every accepted pair from this batch to these claims. The first positive comparison therefore has the earliest

possible timestamp and the smallest eligible witness identifier. Recording it establishes the exhaustive boundary and witness. Removing the resolved claim is safe because subsequent batches have later timestamps.

After processing all comparisons, insert the batch's claims. They cannot have a strictly later witness among previously processed records. They therefore satisfy the unresolved invariant. Induction establishes the output for every batch. Any remaining claim has no accepted later witness and receives an infinite boundary.

For the supplied positive frame threshold, an accepted pair with nonempty frames shares at least one token. The union of frame-token postings therefore contains every such candidate. Empty frames use a common posting. Repeated posting visits are deduplicated before the gate test. For other gate definitions, completeness must be proved separately or ensured through a full unresolved scan.

B Source and Scoring Details

The complete TempLAMA audit recovers all 300 keys used by the released adaptation. It checks 2,677 original yearly records. There are 613 multi-answer records and 245 keys with at least one such record. The original builder takes the first answer identifier and collapses consecutive repetitions. Rebuilding that procedure exactly reproduces the released claim texts and answer identifiers. The audit finds 153 transitions where the previous selected answer remains in the next complete answer set.

Our source-set evaluation accepts all answer identifiers from the latest source observation at the cutoff. It retains the existing claim pool and every method's previously computed ranking. No parameter changes follow this rescore. The old selected-answer oracle is not a complete source-set oracle. It is therefore excluded from primary source-set superiority claims.

The single-answer subset contains keys with exactly one answer identifier in every source year. This criterion selects 55 keys before examining retrieval outcomes. The experiment retains all 747 claims as candidates. Its 57 historical probes and 55 current probes therefore preserve the original distractor pool.

Historical cluster intervals resample keys and retain every sampled key's probes. We use 5,000 resamples with seed 20261006. This preserves dependence between dates from the same chronology.

Corpus	n	Flat	Original mask
Wikidata	33	42.42	96.97
TempLAMA	300	56.67	93.67
Wikipedia prose	35	71.43	85.71

Table 6: Archived Qwen contained-value accuracy, in percent. These runs use the original single-target labels and ranking. They are inherited results, not fresh model evaluations.

All method comparisons use aligned questions and cutoffs. The saved prediction records contain selected identifiers, target identifiers, and per-query metrics.

C Archived Current-Answer Runs

The inherited reader uses Qwen2.5-3B-Instruct, greedy decoding, and at most 24 new tokens. Its contexts contain five claims from a hybrid lexical and dense ranking. The dense encoder is all-MiniLM-L6-v2 (Reimers and Gurevych, 2019). Prompts include the available dates and request one answer value. These settings and generated outputs precede the certificate study. The compiler checks establish identical current masks under the unchanged pair gate. Thus the archived contexts remain applicable to the current path. They do not measure new historical reader performance.

The archived scorer checks whether the normalized selected answer appears in the generated response. We call this *contained-value accuracy*. A stricter version also rejects responses containing a competing archived answer. Neither metric is normalized whole-answer exact match. The TempLAMA rows use the original selected-answer transition task. They do not use the complete source-answer scoring introduced in Section 5.

On that task, archived lifecycle selection reaches 0.937 contained-value accuracy across 300 TempLAMA questions. Flat retrieval reaches 0.567 and the dated-prompt variant reaches 0.623. On 33 Wikidata questions, the corresponding lifecycle and flat results are 0.970 and 0.424. On 35 RealProse questions, they are 0.857 and 0.714. The RealProse paired difference has archived interval $[0.000, 0.286]$ and reported $p = 0.076$. These results describe the previous current-answer system.

The archived temporal-score comparisons adapt TempRALM and half-life formulas to the same local encoder. They do not reproduce those papers’

complete systems. The selected half-life setting uses nine Wikidata development keys. Those keys remain in the 33-key archived aggregate. The full result cards preserve that selection history.

D Implementation and Reproduction

The released compiler uses the Python standard library and the supplied feature parser. Its output records claim identifiers, timestamps, witness identifiers, source hashes, and detector settings. It also records pair tests and posting visits. An infinite endpoint is encoded as a null endpoint in JSON. An undated start has no historical eligibility and remains separately identifiable.

The correctness runner compares indexed output with an exhaustive gate scan. It checks repeated values, shuffled inputs, equal timestamps, missing timestamps, and arbitrary pair relations. Constructed cases are labeled as software checks. Benchmark results come from the supplied source collections. The run manifest distinguishes new CPU experiments from inherited model generations. The experiment scripts preserve per-query and per-intervention outputs alongside aggregate tables.

Each historical certificate identifies the source that caused exclusion. Each evaluation record identifies the claims actually selected. These records support inspection of an interval error separately from a ranking or reader error.

E AI Assistance

AI tools assisted with literature searches, code development, proof drafts, and manuscript editing. No new language-model inference contributed to the revised experimental tables.